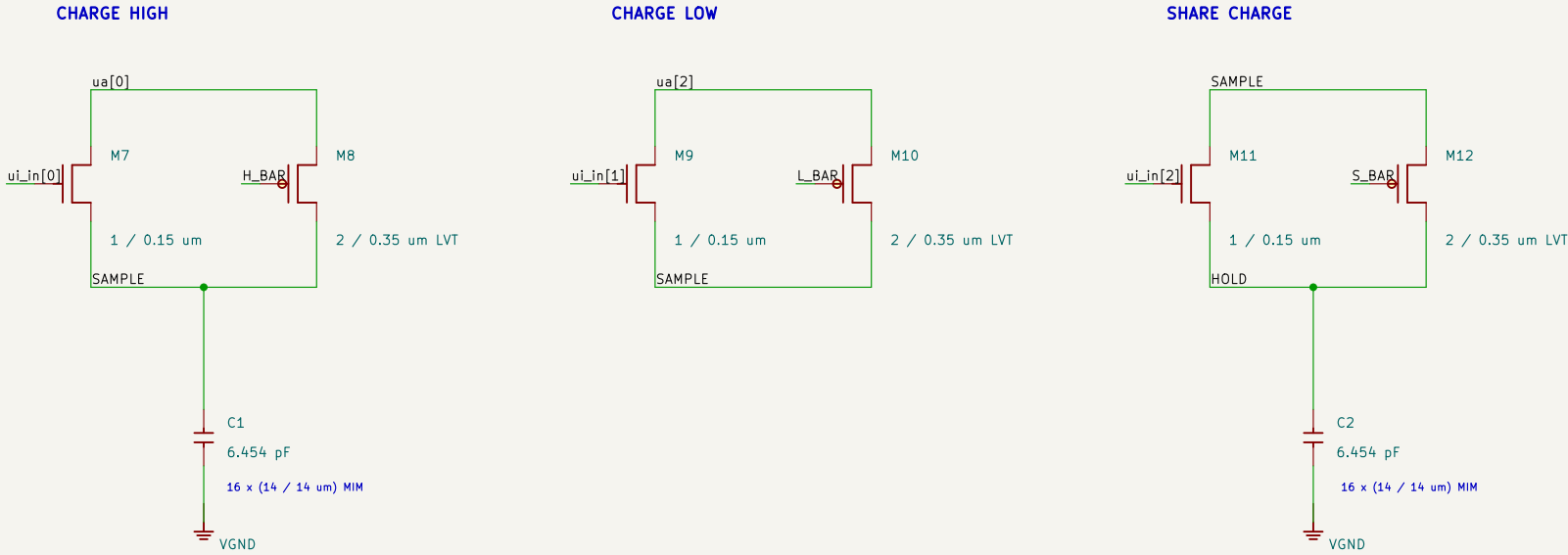


SUAREZ TWO-CAPACITOR DAC / SKY130

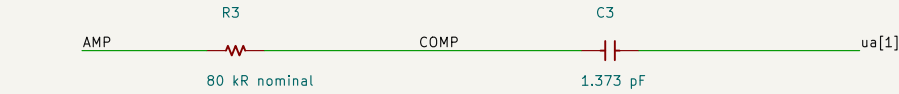
Tiny Tapeout 1x2 analog macro | 1.8 V | LSB first | transistor dimensions shown as W / L

01 CHARGE REDISTRIBUTION



HIGH and LOW must never overlap. SHARE closes only after HIGH / LOW have opened.

02 BUFFER COMPENSATION

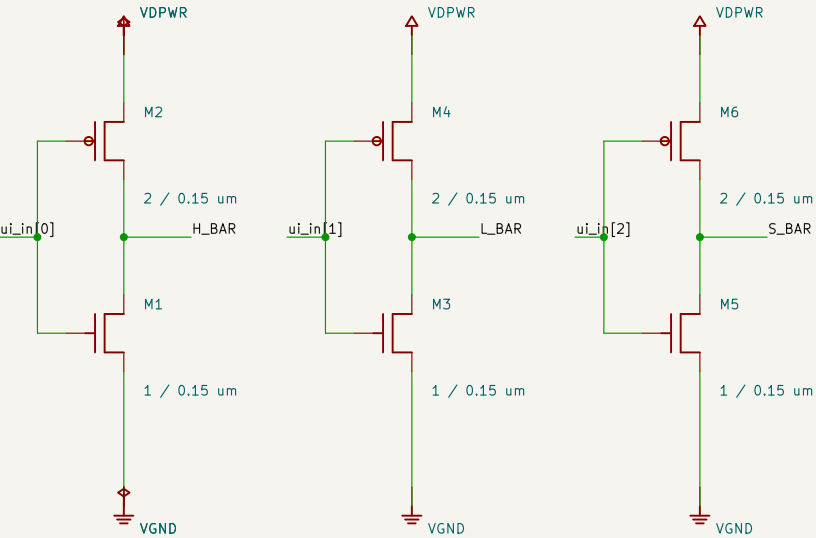


R3: xhigh poly, 0.35 / 14 um

C3 stabilizes the output buffer.

C1 and C2 perform the D/A conversion.

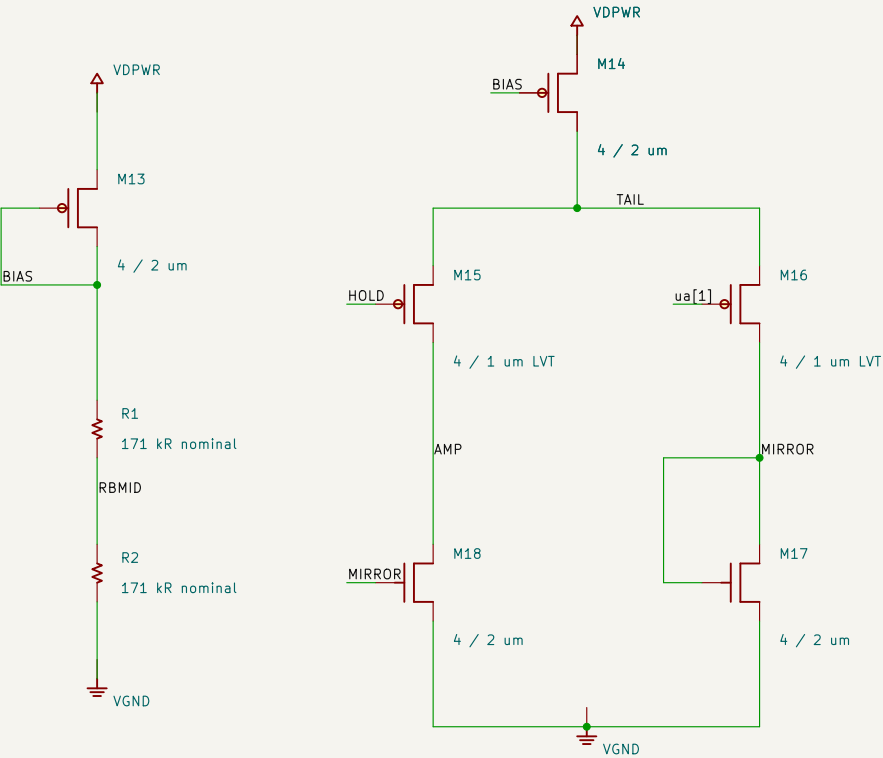
03 COMPLEMENTARY SWITCH CONTROLS



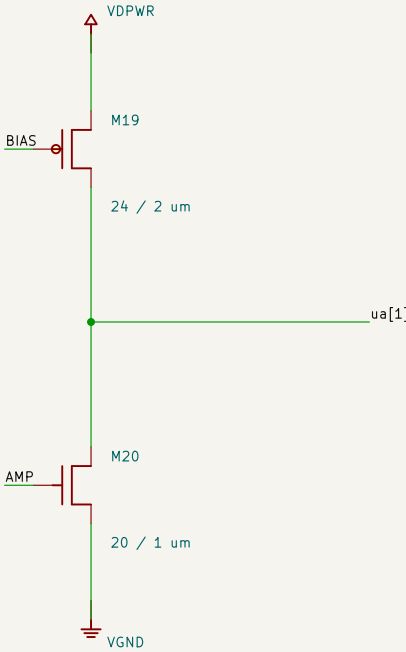
External phases provide dead time.

No on-chip word counter is required.

04 INPUT AMPLIFIER AND BIAS



05 OUTPUT DRIVER



Buffered analog output

PORTS

ui_in[0] = HIGH ui_in[1] = LOW ui_in[2] = SHARE ua[0] = VREFH ua[1] = VOUT ua[2] = VREFL

Initialize: HIGH=0, LOW=1, SHARE=1 for 8 us. Then 8 bits, LSB first: charge 2 us, dead 0.2 us, share 2 us, dead 0.2 us.

Bulk ties: every NMOS -> VGND; every PMOS -> VDPWR. Bulk pins are implicit in this drawing and explicit in the silicon netlist.

Unused digital outputs are tied to VGND; ua[3:7] are isolated. Supply bypassing is provided by the Tiny Tapeout chip / board.

SKY130 / Tiny Tapeout

Sheet: /
File: suarez_dac.kicad_sch

Title: **Suarez two-capacitor DAC**

Size: A3

Date:

Rev: 0.1

KiCad E.D.A. 10.0.6

Id: 1/1